ME-2 Digital Twin Baseline Construction

Weeks 1–2: System Onboarding & Scope Alignment

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1 Purpose

The purpose of Weeks 1–2 is to re-establish the role of the ME-2 digital twin within the larger cyber-physical bridge monitoring system and align the Fall work with the models, assumptions, and experimental framework developed during the Spring semester.

During the Spring, the ME-2 effort focused on developing a behavioral digital twin of the instrumented bridge structure. This included sensor calibration, static-response modeling, dynamic vibration characterization, environmental compensation, and definition of residual-based outputs that could eventually be passed to the CpE AI pipeline.

The Fall system extends that work by introducing intentional cyber and cyber-physical attack conditions. Because of this, it is important to clearly separate the responsibilities of the mechanical digital twin from those of the cybersecurity and AI subsystems.

The ME-2 role for the Fall semester is therefore defined as:

Physical validation and behavioral reference, not cyber modeling

The digital twin should determine whether the observed physical response of the structure is consistent with its expected behavior. The CpE systems can then use this information alongside cyber, timing, and telemetry features to determine whether an attack or anomaly is present.

2 Spring Digital Twin Handoff

The Spring work established the initial physical baseline needed for Fall integration. The digital twin was developed primarily as a behavioral model rather than a high-fidelity finite element representation of the structure.

For a measured structural response y, the basic twin relationship is:

$$r(t) = y_{\text{measured}}(t) - y_{\text{expected}}(t)$$

where $r(t)$ is the residual between the measured response and the response predicted by the baseline model.

The Spring implementation established several types of physical reference information:

- Static load-response relationships
- Expected sensor operating ranges
- Statistical tolerance bounds
- Residual error signals
- RMS acceleration and vibration magnitude
- Dominant vibration frequency and FFT features
- Temperature-conditioned response compensation

These outputs provide the physical reference layer needed for Fall attack testing.

The MATLAB models developed during the Spring semester can therefore be reused rather than rebuilding the twin from the beginning. The Fall effort will focus on validating and extending those models as new attack datasets become available.

3 Relationship Between Attacks, Twin, and AI

The Fall architecture can be viewed as three interacting layers:

$$\text{Attack / Telemetry Layer} \rightarrow \text{Physical Digital Twin} \rightarrow \text{AI Detection Layer}$$

The CpE attack and telemetry work introduces labeled normal and abnormal operating conditions into the system. These conditions may include manipulated sensor measurements, timing disturbances, replayed data, or other changes to the information being transmitted through the monitoring system.

The ME digital twin does not need to determine how the attack was generated. Instead, it evaluates the physical consequence of the received data.

For example, a sensor value may appear valid from a communication standpoint but still be inconsistent with the expected structural response. The twin can identify this disagreement through an abnormal residual:

$$|r(t)| > r_{\text{allowed}}$$

The AI pipeline can then combine the twin output with the cybersecurity features generated by the CpE teams.

Conceptually, the AI model can receive:

$$\mathbf{x}_{AI} = [\mathbf{x}_{cyber}, \mathbf{x}_{timing}, \mathbf{x}_{physical}]$$

where $\mathbf{x}_{physical}$ contains features generated by the ME digital twin.

Examples include residual magnitude, normalized residual, bound violations, acceleration RMS, and frequency shift.

This separation keeps each subsystem focused on its intended role while still allowing the final system to combine information from multiple domains.

4 ME-2 Role for Fall 2026

The primary responsibility of ME-2 is to determine whether incoming measurements remain physically consistent with the known behavior of the bridge.

This includes three main functions.

First, the existing MATLAB models provide expected values for the measured response. For a simple static condition, the baseline relationship may take the form:

$$\hat{y} = ax + b$$

while environmental conditioning can extend the model to:

$$\hat{y} = b_0 + b_1 L + b_2(T - T_{\text{ref}})$$

Second, the twin determines whether the measured response remains inside its expected operating envelope:

$$y_{\text{lower}} \leq y_{\text{measured}} \leq y_{\text{upper}}$$

Finally, the twin generates deviation metrics that can be passed to the CpE AI models instead of requiring those models to infer physical behavior directly from raw sensor values.

The ME-2 effort will therefore concentrate on the physical meaning of the data rather than implementing attack logic, encryption, network-security models, or cyberattack classification.

5 Physical Assumptions to be Checked

The Spring models were developed using a set of baseline assumptions that must be rechecked before they are applied to Fall attack datasets.

Table 1: Physical Assumptions for Fall Validation

Assumption	**Fall Validation Requirement**
Sensor calibration remains valid	Confirm load-cell and accelerometer calibration has not changed significantly since the Spring testing period.
Sensor orientation is unchanged	Verify accelerometer orientation and sign convention before comparing Fall measurements with Spring baselines.
Structural configuration is unchanged	Confirm beam geometry, supports, sensor locations, and vibration motor placement remain consistent with the modeled configuration.
Static response remains approximately linear	Repeat selected loading conditions to verify that the existing static linear model remains representative.
Sampling rate is known and stable	Confirm timestamp spacing before calculating FFT or other time-dependent features.
Dynamic baseline remains repeatable	Compare vibration RMS and dominant frequency with previous baseline behavior before using frequency shift as an anomaly feature.
Temperature affects sensor response predictably	Use temperature as a conditioning variable where available rather than treating thermal drift as structural deviation.
Sensor channels remain synchronized	Confirm load, acceleration, vibration, and environmental measurements represent the same physical time window.
Baseline data represent healthy behavior	Verify that the reference dataset contains no known attack, hardware fault, or abnormal structural condition.

These checks are important because an incorrect physical baseline would produce misleading residuals and could cause the AI system to learn the wrong distinction between normal and abnormal behavior.

6 Interface with Fall AI Work

The Fall AI pipeline should not require direct knowledge of the mechanical model equations. Instead, ME-2 can provide a compact set of physically meaningful features derived from the twin.

A typical output vector may include:

$$\mathbf{x}_{physical} = [r,\ |r|,\ r/\sigma,\ \text{bound flag},\ a_{\text{RMS}},\ f_{\text{dom}},\ \Delta f]$$

This provides the AI model with information describing *how physically unusual* the current response is.

The attack label remains a CpE-side quantity. The ME twin should not automatically equate a large residual with a cyberattack because a physical fault, environmental change, sensor failure, or unexpected loading condition could produce similar behavior.

Instead:

$$\text{Twin Output} = \text{Physical Deviation}$$

while:

$$\text{AI/CpE Output} = \text{Attack or Anomaly Classification}$$

Maintaining this distinction will make the final system easier to interpret and validate.

7 Weeks 1–2 Outcome

The primary outcome of the re-onboarding period is a clarified division of responsibility between the mechanical and computer engineering portions of the project.

The Spring ME-2 work already provides the foundation for expected-value modeling, operating bounds, residual generation, vibration features, and environmental compensation. Fall development can therefore focus on checking these physical assumptions against the current hardware configuration and applying the twin to the labeled datasets produced by the CpE teams.

The ME-2 digital twin will serve as the physical validation layer of the system. Its role is not to identify the implementation details of an attack, but to determine whether the behavior represented by the incoming data remains consistent with the known physics and experimentally established baseline of the bridge.